

MAE 140 Winter 12 Homework #2

T R & T, 2.41, 2.42, 2.44, 2.50, 2.52, 2.53, 2.58, 3.36, 3.39, 3.41

Due: 01/24/2012

2-41 Use voltage division in Figure P2-41 to obtain an expression for v_L in terms of R , R_L , and v_S .

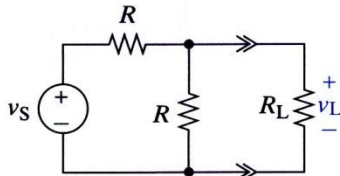


FIGURE P2-41

2-42 Use current division in Figure P2-42 to find v_L if $R = 10\ \Omega$, $R_L = 20\ \Omega$, and $i_S = 2.5\text{ mA}$.

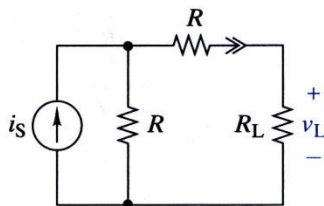


FIGURE P2-42

2-44 The $1\text{-k}\Omega$ load in Figure P2-44 needs 5 V across it to operate correctly. Where should the wiper on the potentiometer be set (R_X) to obtain the desired output voltage?

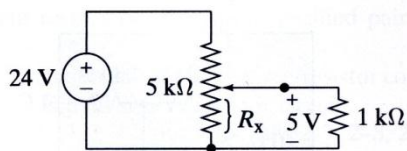


FIGURE P2-44

2-50 Select a value of R_X in Figure P2-50 so that $v_L = 2\text{ V}$. Repeat for 4 V and 6 V . Caution: R_X must be positive.

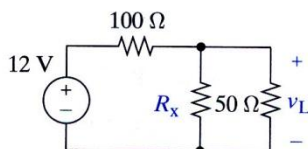


FIGURE P2-50

2-52 Use circuit reduction to find v_x and i_x in Figure P2-52.

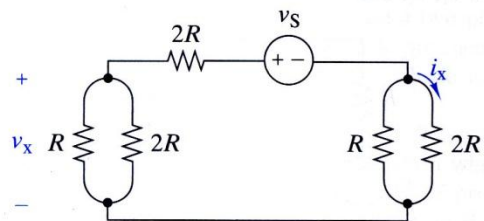


FIGURE P2-52

2-53 Use circuit reduction to find v_x and the power provided by the source in Figure P2-53.

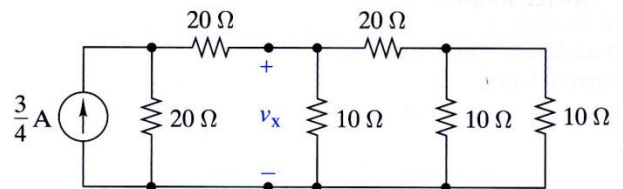


FIGURE P2-53

2-58 The current through R_L in Figure P2-58 is 40 mA . Use source transformations to find R_L .

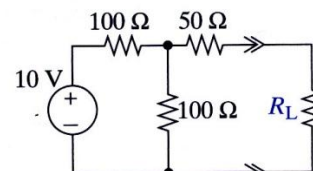


FIGURE P2-58

- 3-36(a)** Find the Thévenin or Norton equivalent circuit seen by R_L in Figure P3-36.
(b) Use the equivalent circuit found in part (a) to find i_L in terms of i_S , R_1 , R_2 , and R_L .
(c) Check your answer to part (b) using current division.

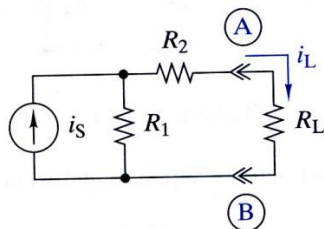


FIGURE P3-36

- 3-39** Find the Thévenin equivalent seen by R_L in Figure P3-39. Find the power delivered to the load when $R_L = 50 \text{ k}\Omega$ and $200 \text{ k}\Omega$.

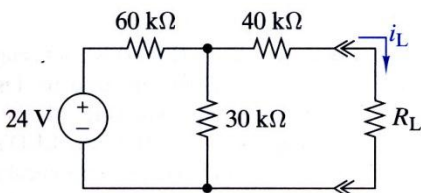


FIGURE P3-39

- 3-41** The purpose of this problem is to use Thévenin equivalent circuits to find the voltage v_x in Figure P3-41. Find the Thévenin equivalent circuit to the left of terminals A and B.

Find the Thévenin equivalent circuit to the right of terminals A and B. Connect these equivalent circuits together and find the voltage v_x .

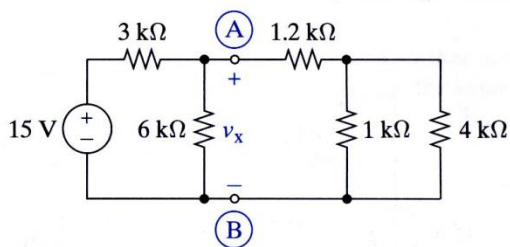


FIGURE P3-41